



The Bruhat–Tits Tree Code: What's Verified, What We Plan to Test Next

Rowan Brad Quni-Gudzinás — QNFO · CWI Summer School on Quantum Algorithms & QEC · Poster session Wed 26 Aug 2026, 16:30–18:00

1 · The code in one line

Holographic **perfect-tensor codes on a truncated Bruhat–Tits tree**: a $[[3,1,1]]$ perfect tensor sits at every internal vertex, logical information lives at the root, physical qubits at the leaves, and error correction is built into the tree geometry rather than applied by active measurement loops.

Error model	Tree code	Surface code	Ratio
Bit-flip	50.0%	~10.9%	4.6×
Depolarizing	75.0%	~1.0%	75×
Independent X+Z	17.30%	—	—

Computed three independent ways — analytical recursion, exact Gottesman–Knill stabilizer simulation, Monte Carlo — and re-verified by a deposited 35-check script.

3 · Test plan (near-term first)

#	Test	Where	Disconfirmation criterion
T1	Matched-noise threshold comparison: tree vs surface under the <i>same</i> correlated noise model	Classical	Tree advantage falls to parity under matched noise
T2	Decoder complexity + full overhead (physical qubits + classical cost)	Classical	Advantage disappears once decoding cost is included
T3	Return-probability decay of a random walk on a p-regular tree graph	Analog / ion sim, ~8 wk	No change in decay exponent at the predicted value
T4	Clock-rest ultrametricity split: diagonal 0% vs non-diagonal 29–35%	Trapped ion, ~8 wk	Measured split outside the predicted range
T5	Independent re-implementation of T1 by an external group	Community	Reproducibility test, not a prediction

T1 and T2 are pure classical work — cheap, decisive, and doable immediately; they answer the two objections this room will raise first. Every criterion is numeric, so a null result is publishable.

2 · Verification status (as of 21 Aug 2026)

Classical validation: done, three methods. **Hardware: none yet** — no tree code has run on a quantum processor; everything above is simulation.

First falsifiable prediction — tested, null. Log-periodic oscillations in the Planck 2018 temperature spectrum: **no signal found** at any resolvable radix (bootstrap $p = 0.89$); bispectrum: upper bounds only. The testing machinery runs and publishes nulls.

Published boundary. Under *independent* errors the qudit family sits at $\approx 2 \times 10^{-4}$ (~55× below surface codes) — the architectural claim targets the *correlated-failure* regime, and that boundary is published, not hidden.

No-go engagement. The QEC–Darwinism tradeoff ($F_L > 0.874$) was audited on tree spaces: collective fragment refinement moves the boundary to $F_L \approx 0.83$ –0.85, residual gap disclosed.

Self-correction published. A naive p-adic valuation reading of stabilizer codes carries no new content; the program published the correction rather than the confirmation.

4 · What we ask of this room

1. An openly specified **correlated-noise benchmark** (data + model) we can run T1 against.
2. Decoder reality check: does a known decoder family already cover tree-structured codes, and how does its complexity scale?
3. Pointers on **perfect tensors for $p > 2$** (binary $[[3,1,1]]$ is settled; larger primes are open).
4. **Independent re-run of T1** — we hand over the simulator and the protocol; a third-party rerun settles the classical claim.

Horizons

2026: T1–T2 + open simulator release + ZX gate compilation · **2026–27:** T3–T4 instrumentation, noise-model publication · **Long term:** prototype path only if matched-noise scrutiny survives; energy accounting (joules per correct answer) as the mission-level comparison.

References (DOIs verified 21 Aug 2026): Tree-code architecture 10.5281/zenodo.20109836 · Guide + 35-check verification script 10.5281/zenodo.22038733 · Planck 2018 null 10.5281/zenodo.21902891 · CMB bispectrum bounds 10.5281/zenodo.21901664 · Qudit QEC 10.5281/zenodo.21046993 · Trapped-ion testbed 10.5281/zenodo.22025544 · QEC–Darwinism audit 10.5281/zenodo.21964674 · Prime-valuation correction 10.5281/zenodo.21979060 · ZX seam 10.5281/zenodo.21992118 · **v5 candidate for the poster decision of Mon 24 Aug** — print A0 portrait, margins none, scale 100%.

